CASE STUDY

HighProbTrade — AI Stock Analysis Platform

Type: Client Project

Domain: Fintech · AI

Status: 🟡 Testing Phase — Pre-Launch

Client Project | Fintech · AI | Product Portfolio | HighProbTrade

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At a Glance

ATTRIBUTE	DETAIL
Client	HighProbTrade
My Role	Solo product owner + full-stack builder (vibe coding)
Stack	React 19 · Vite · Firebase · Gemini AI Studio · Tailwind CSS
Build Method	Vibe coding — Claude + Antigravity
Deployment	Vercel (pre-launch)
Users	Retail NSE F&O traders

Problem Statement

Retail traders participating in NSE Futures & Options face a specific information problem: **too much data, too little signal.**

Existing tools gave traders raw option chain data with no interpretation. To find a tradeable setup, a retail trader had to:

- Manually scan 188+ F&O stocks
- Cross-reference Open Interest changes
- Assess market sentiment independently
- Identify intraday breakout setups
- Do all of this before the trading window closed

The gap: **no tool combined AI-driven sentiment, real-time OI tracking, and multi-factor filtering into a single actionable interface for retail traders.**

User Research

Conducted interviews with retail F&O traders to understand their current workflow:

Pain points identified:

- Information overload — too many stocks, too many signals
- Delayed data — most free tools had 15-minute delays
- No context — raw OI numbers without interpretation
- No prioritisation — traders wasted time on low-probability setups

User persona:

> *Retail trader, 25–40 years, trading NSE F&O part-time or full-time, using Zerodha or Upstox, checking 3–5 platforms before placing a trade. Wants: fewer, better setups. Not more data.*

Market Analysis

SEGMENT	SIZE	OPPORTUNITY
Active NSE F&O retail traders	~5M registered	High engagement, unmet need
Existing tools (Sensibull, Opstra)	Institutional-grade, complex	Retail UX gap
AI-first trading tools	Emerging category	First-mover advantage possible

Solution: 4-Phase Smart Intraday Scanner

Designed a structured scanning methodology that narrows from 188 stocks to 2–6 high-quality setups:

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Phase 1: Market Macro View
→ Live quotes for all 188 F&O stocks
→ Advances vs Declines → Bullish / Bearish / Sideways signal
→ A/D bar visualisation

Phase 2: Sector Heatmap
→ 14 named sectors, ranked by average % change
→ Bullish: Top 2 green sectors selected
→ Bearish: Bottom 2 red sectors selected

Phase 3: PDH/PDL Master Filter
→ 7-day daily chart to extract Previous Day High/Low
→ Bullish: keeps only price > PDH
→ Bearish: keeps only price < PDL

Phase 4: Deep Analytics
→ Full OI Chain, PCR, Max Pain, RSI/MACD/EMA
→ AI Insight per stock (Gemini AI Studio)
→ 5-level sentiment: Strongly Bullish → Strongly Bearish
```

Features Built

Nifty OI Tracker

- Polls options chain every 15 minutes
- Tracks Total Call OI and Total Put OI in Lakhs
- Calculates 15-minute delta for immediate sentiment
- Displays PCR, underlying value, and Bullish/Bearish flag

Intraday Breakout Detection

- 15-minute candles over rolling 5-day span
- First 30-min High/Low range established from first two candles
- Breakout signal fires if latest candle closes beyond the 30-min range
- Signal types: `30MIN HIGH BREAK` or `30MIN LOW BREAK`

OI Visualisation

- 5 ITM + ATM + 5 OTM focus (11 most relevant strikes)
- Support = Highest Put OI in visible window
- Resistance = Highest Call OI in visible window
- Inline horizontal OI bars proportional to OI and change-in-OI

RICE Prioritisation

FEATURE	REACH	IMPACT	CONFIDENCE	EFFORT	PRIORITY
4-phase scanner	High	High	High	High	P0
Real-time Nifty OI tracker	High	High	High	Medium	P0
AI stock insights (Gemini)	High	High	Medium	Medium	P1
OI visualisation with S/R	Medium	High	High	Medium	P1
Lead capture website	High	Medium	High	Low	P2

Build Approach

Built entirely using vibe coding — solo, from product planning to deployment:

- **Planning:** PRD, user stories, acceptance criteria authored first
- **Frontend:** React 19 + Vite + Tailwind CSS 4
- **Backend:** Firebase (Firestore + Auth) + Express + Socket.io (real-time)
- **AI layer:** Gemini AI Studio for per-stock trade insights
- **Market data:** `stock-nse-india` package (resolved 403 issues from manual axios fetching)
- **Deployment:** Vercel

Key Technical Challenges Solved

Problem: NSE API returned two different expiry date formats (`"21-Apr-2026"` vs `"21-04-2026"`) causing all OI aggregation to return zero.

Fix: Filter now uses `r.expiryDates` (plural field) matching `records.expiryDates[]` — not the per-row CE/PE field.

Problem: `/api/intraday-scan` HTTP route was timing out on large scans.

Fix: Route now returns `{status:'scanning'}` immediately and pushes results via Socket.io `smartScanUpdate` — non-blocking architecture.

Status & Next Steps

- Currently in client testing phase with real trader users
- Gathering feedback on signal accuracy and UX clarity
- Payment integration (₹4,999 / ₹9,999 plans) pending before commercial launch

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